

COMPUTER VISION SYSTEM
Enabling machines to perceive, understand and make decisions from visual information

Built on

1. SYSTEM CAPABILITIES
The foundation that determines what can be seen and how well it can be understood

Leads to

2. IMAGE FORMATION & ACQUISITION
From real world to digital representation

Guided by

3. LITERATURE PERSPECTIVES
Research foundations driving advances

Influence

Provides

4. VISION-BASED APPLICATION SELECTION
Different applications require specific capabilities and performance

Gonzalez & Woods (Digital Image Processing)
• Fundamentals of image processing
• Filtering, Enhancement
• Geometric Operations

Szeliski (Computer Vision)
• 3D Vision
• Multiple View Geometry
• Image Formation Models

Deep Learning (CNNs, Transformers)
• Feature Learning
• Large-Scale Data
• End-to-End Learning
• State-of-the-Art Performance

Operates through

5. COMPUTER VISION PROCESSING LEVELS
Steps from raw data to actionable information

- Image Acquisition**
• Capture visual data
• Convert to digital
- Image Processing**
• Enhancement
• Filtering
• Segmentation
- Feature Analysis**
• Extract meaningful features
• Detect patterns
- Decision Making**
• Classification
• Recognition
• Action / Output

Supported by

AI / Machine Learning
Learns from data to improve performance and generalize to new scenarios

Classification & Recognition

Transforms extracted information into meaningful decisions

Image Formation

Lighting Conditions
• Illumination
• Contrast
• Ambient Light

Sampling
• Spatial Resolution
• Sampling Rate
• Aliasing Control

Quantization
• Intensity Resolution
• Bit Depth
• Gray Levels

Enhances

6. SYSTEM PERFORMANCE & EFFECTIVENESS
Ensures the computer vision system delivers value in real-world applications

Accuracy
Correct detection and reliable results

Speed
Real-time or near real-time processing

Reliability
Consistent performance in varying conditions

Robustness
Works in noise, occlusion and changing environments

Improves

Real-Time Intelligent Applications
Improving decisions, enhancing safety and creating better future